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New distribution record of *Anopheles rivulorum*-like from Sadiola, Mali, with notes on malaria vector insecticide resistance

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Background: The SEMOS gold mine in Sadiola, southwestern Mali, has been implementing a malaria vector control programme for 15 y using indoor residual house spraying and sporadic larval control. Periodic screening of the vector populations have been carried out over the years to provide information to the control programme, mainly on vector species present and their insecticide resistance status. The data from five entomological surveys, carried out in 2006, 2011, 2014, 2016 and 2018, are presented.

Methods: Adult mosquitoes were collected resting on walls inside houses and on verandas. Insecticide susceptibility assays were carried out and mosquitoes subsequently identified by species using molecular assays.

Results: The major malaria vector mosquitoes, *Anopheles gambiae* and *Anopheles arabiensis* were abundant at each sampling period with *Anopheles coluzzii* and *Anopheles funestus* being rare or absent. *Anopheles rivulorum* was identified in 2006 and *Anopheles leesoni* in 2016. The presence of *Anopheles rivulorum*-like, identified for the first time in 2018, was not screened for in previous surveys. Insecticide susceptibility bioassays showed resistance in both *A. gambiae* and *A. arabiensis* to pyrethroids, carbamates and dichlorodiphenyltrichloroethane over the 12 y.

Conclusions: This is the first record of *A. rivulorum*-like west of Côte d'Ivoire. Resistance levels to the three classes of insecticides were variable but appeared to decrease after pyrethroids were discontinued for house spraying.

Keywords: *Anopheles funestus*, *Anopheles gambiae* complex, *Anopheles rivulorum*-like, insecticide resistance, Mali, Sadiola

Introduction

Malaria transmission in Mali occurs in the central and southern areas of the country and in 2018 there were an estimated 3.7 million cases reported in the World Health Organization's (WHO's) annual World Malaria Report.[1] Transmission prevention measures focus mainly on the distribution of long-lasting insecticide-treated bed nets with targeted indoor residual house spraying (IRS) occurring in some areas. Malaria vector control programmes, whether they implement LLINs or IRS, require regular monitoring of the vector populations. This includes species diversity, behaviour and most importantly, insecticide resistance status of the mosquito populations present in the control areas. The major vectors in Mali belong to the *Anopheles gambiae* complex and the *Anopheles funestus* group.[2] Insecticide resistance in members of the *A. gambiae* complex has been widely recorded in both southern and central Mali.[3,4] The Sadiola SEMOS gold mine, situated in the Kayes district of southwest Mali near the border with Senegal (Fig. 1), implemented a malaria vector control programme in 2005 as a community outreach initiative to help reduce the incidence and mortality of malaria in mine employees and surrounding communities. There are 17 villages surrounding the mine that are incorporated into the malaria vector control programme, which is mainly based on IRS. Entomological surveys were carried out in 2006, 2011 and 2014 and based on these results, the insecticides used for IRS have been changed over the years to manage the emerging insecticide resistance.[5]

There are four main malaria vectors in the Sadiola area: *Anopheles coluzzii*, *A. gambiae*, *Anopheles arabiensis* (all members of the *A. gambiae* complex) and *A. funestus*.[5–8] *A. funestus* belongs to a group of morphologically similar species[6] that includes minor vectors such as *Anopheles rivulorum* and *Anopheles leesoni* (Tanzania[9,10]) and more recently *Anopheles vaneedeni* and *Anopheles parensis* (South Africa[11,12]). This article presents

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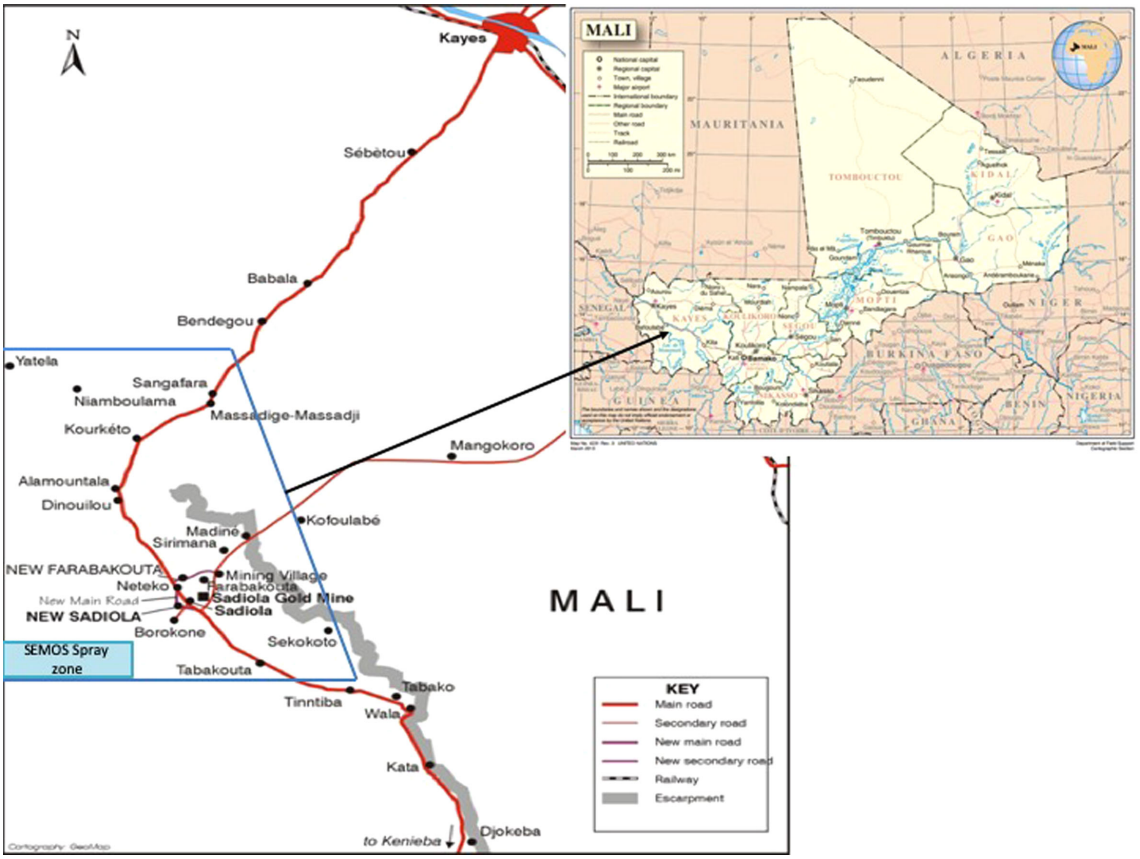


Figure 1. Map of the Sadiola area in the Kayes district of southwest Mali. The spray zone of the SEMOS vector control programme is outlined in blue. Collections were carried out in villages within this zone as well as villages along the main road north and south of the sprayed area.

an update on the vector species groups at the SEMOS gold mine[5] and records for the first time west of Côte d'Ivoire the presence of *A. rivulorum*-like in Mali. Insecticide susceptibility results over the five surveys are summarized.

Materials and methods

Study site

Mali is made up of nine administrative regions: Bamako (the capital), Goa, Kayes, Kidal, Koulikoro, Mopti, Ségou, Sikasso and Tombouctou. The Kayes administrative region is in the Sudano-Sahelian ecological zone, in the southwest of the country, bordering on Senegal (Fig. 1). The SEMOS gold mine is situated in Sadiola (13°53′N, 11°42′W), 70 km south of the town of Kayes. Malaria is holo-endemic, peaking during the rainy season of June–October.

Field collections

Entomological surveys were conducted in 2006, 2011 and 2014 in Sadiola both inside and outside the spray zone and results presented in Wragge et al.[5] The 2016 and 2018 surveys were conducted in the unsprayed villages of Babala and Bendegou just north of the spray zone (Fig. 1) in order to maximise the number of mosquitoes collected. All surveys were conducted towards the end of the rainy season (Table 1). Using hand aspirators, resting adult mosquitoes were collected indoors and on veranda areas and roofs covered with palm leaves.

Species identification, parasite infection and insecticide susceptibility bioassays

Samples were identified using morphological keys for anopheline species[6,7] and after bioassays were conducted, they were preserved individually on silica gel in Eppendorf tubes. Further molecular identifications were carried out for the *A. gambiae* complex[13,14] and *A. funestus* group.[15,16] Mosquitoes were screened for *Plasmodium falciparum* infection using the enzyme-linked immunosorbent assay of Wirtz et al.[17] Insecticide susceptibility bioassays were carried out in a field laboratory, except for the 2014 collections, where live adults were transported to the laboratory in Johannesburg for egg-laying. Bioassays were carried out according to the standard protocols of the WHO.[18] Three insecticide classes were tested: pyrethroids (0.05% deltamethrin), carbamates (0.1% bendiocarb) and organochlorines (4% dichlorodiphenyltrichloroethane [DDT]). Pooled wild-caught female mosquitoes of unknown age plus the 3- to 5-day-old F1 progeny of those laying eggs in Johannesburg in 2014 were exposed to treated papers (maximum 25 per tube) for 60 min and

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Table 1. Indoor resting *Anopheles* species in Sadiola identified using polymerase chain reaction methods (2006, 2011 and 2014 data from Wragge et al.[5])

Year	Species identified	Number collected	Total per year, %
2006 September–October	*A. gambiae*	6	3.4
	A. coluzzii	6	3.4
	A. arabiensis	147	83.1
	A. funestus	11	6.2
	A. rivulorum	7	3.9
		Total=177	
2011 July–August	*A. gambiae*	196	77.2
	A. coluzzii	5	2.0
	A. arabiensis	43	16.9
	A. funestus	10	3.9
		Total=254	
2014 August	*A. gambiae*	165	69.6
	A. coluzzii	15	6.3
	A. arabiensis	57	24.1
		Total=237	
2016 August	*A. gambiae*	261	53.0
	A. coluzzii	18	3.6
	A. arabiensis	182	37.0
	A. funestus	16	3.3
	A. leesoni	15	3.1
		Total=492	
2018 September	*A. gambiae*	60	7.4
	A. coluzzii	15	1.9
	A. arabiensis	716	88.5
	A. funestus	1	0.1
	A. rivulorum-like	17	2.1
		Total=809	

mortality recorded 24 h later. Controls were exposed to untreated papers, with mortality never exceeding 2%. The WHO criteria are as follows: >98% mortality is susceptible, 90–97% mortality is suspected resistance requiring additional tests and <90% mortality is confirmed resistance.[18]

Results and discussion

The entomological surveys were carried out to provide the SEMOS vector control programme with specific information to guide intervention policy and, as such, the only constant variables throughout the five surveys were the mosquito collection method—indoor resting adults—and the insecticide bioassay method. As a result, very little can be concluded from the results on, for example, changes in vector density over time. Table 1 presents the results of the mosquito collections for the five surveys with data from 2006, 2011 and 2014 coming from Wragge et al.[5] While the abundance of *A. gambiae* and *A. arabiensis* varied at different time points, these were always the most abundant species with *A. coluzzii* always in the minority (1.9–6.3%). This is similar to results presented for the Kayes district by Sovi et al.,[4] where *A. gambiae* was dominant in 2017 and *A. arabiensis* was dominant in 2018.[4] Members of the *A. funestus* group were captured in very small numbers, with 31 collected in 2016, only 18 in 2006 and 2018, 10 in 2011 and none in 2014. *A. rivulorum* was found only once in 2006 and *A. leesoni* once in 2016. The molecular assay used for identification of the *A. funestus* group[15] did not include primers for detecting *A. rivulorum*-like until 2018[16] and it is therefore not known if this species was present previously.

Very little is known about *A. rivulorum*-like. Hackett et al.[19] first described a possible new member of the *A. funestus* group in a study carried out in Burkina Faso, with comparison samples from Kenya and South Africa. Studies carried out in Cameroon and South Africa again highlighted the existence of a new species within the *A. funestus* group, giving it the name *A. rivulorum*-like.[20] Subsequently *A. rivulorum*-like was identified in northeastern Côte d'Ivoire,[21] Zambia,[22] Malawi[23] and South Africa.[16] Assays for parasite infections have found no *P. falciparum* in the *A. rivulorum*-like tested except for a single specimen in Malawi.[23] Likewise, assays for parasite infection on the 17 wild-caught *A. rivulorum*-like collected in the 2018 Sadiola survey indicated no positives for *P. falciparum*. The potential for *A. rivulorum*-like biting humans was shown in a study in western Burkina Faso, where one

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Table 2. Summarized insecticide susceptibility test results of *A. gambiae* and *A. arabiensis* from five surveys carried out for the SEMOS gold mine vector control programme (2006, 2011 and 2014 data from Wragge et al.[5])

		Mortality at 24 h post-exposure, % (n)		
Year	Species	Deltamethrin	Bendiocarb	DDT
2006	*A. arabiensis*	99.3 (134)	74.0 (81)	94.2 (104)
2014	*A. arabiensis*	85.7 (28)	83.8 (105)	N/A
2016	*A. arabiensis*	86.2 (29)	100 (39)	37.5 (8)
2018	*A. arabiensis*	98.9 (91)	95.0 (80)	86.9 (99)
2011	*A. gambiae*	68.0 (125)	11.9 (59)	38.3 (47)
2014	*A. gambiae*	19.0 (58)	93.6 (109)	2.1 (48)
2016	*A. gambiae*	11.9 (42)	86.3 (51)	7.1 (26)

A. rivulorum-like was caught during human landing sampling.[24] No blood meal analyses were carried out in the Sadiola survey, but the fact that the mosquitoes were collected in human habitations could indicate that they might feed on humans, but this would need to be confirmed. This is the first record of this species in Mali.[25]

The insecticide susceptibility results are presented in Table 2. Apart from 2014, all tests were carried out in the field on wild-caught females. The 2014 susceptibility tests were carried out on the F1 progeny of wild females that had been transported to a laboratory in South Africa.[5] The indoor residual spray programme at the SEMOS mine started using pyrethroids in 2005 and stopped in 2011. The first survey in 2006 found mostly *A. arabiensis* (Table 1) that was fully susceptible to deltamethrin (Table 2). In 2011, *A. gambiae* was the dominant species, with only 68% mortality on pyrethroids. Interestingly, resistance to bendiocarb in both species was high during the implementation of pyrethroids but declined significantly once an organophosphate became the insecticide of choice from 2013 onwards.[5] The 2018 susceptibility of *A. arabiensis* to deltamethrin (98.9%) was higher than that recorded for the Kayes district in 2018 by Sovi et al.,[4] where around 76% mortality was recorded. It is possible that the susceptibility of *A. arabiensis* in Sadiola is due to a lack of selection pressure, with pyrethroids no longer being used for malaria vector control.

Conclusions

A. rivulorum-like was recorded in Sadiola in southwest Mali for the first time. Further studies are needed to determine the insecticide resistance status and the malaria transmission capabilities of this species in Mali. The mosquito vectors *A. gambiae* and *A. arabiensis* from five entomological surveys carried out to guide the SEMOS mine's indoor spray programme over 12 y show considerable variation in both the numbers collected and the resistance to pyrethroids and carbamates.

Authors' contributions: SW and DT implemented the SEMOS mine vector control programme. SW, DT, NV and RH carried out the entomological surveys and laboratory processing of the mosquitoes. SW and MC analysed the data and drafted the manuscript. All authors reviewed and approved the final manuscript.

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Data availability: All data are provided in the article and the statement is provided in the manuscript.

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